Integral optimal flat-norm decompositions in codimension one

Kaiwen Jin
yui.ui.siki@gmail.com

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Abstract

The real flat norm decomposes a current into a residual and the boundary of a filling, minimizing their total mass. We prove that every compactly supported integral d-current in $\mathbb{R}^{d+1}$ has an integral optimal decomposition, including when its boundary is nonzero. The filling is a scalar function of bounded variation. Rounding its values with one common shift preserves the mean filling mass and does not increase the mean residual mass. The integer multiplicities of the input are essential to the latter calculation. A compactness argument then gives an integral optimum.

1 The flat-norm question

Let $\mathcal{N}_k^c(\mathbb{R}^n)$ denote compactly supported real normal k-currents, and let $\mathcal{I}_k^c(\mathbb{R}^n)$ denote compactly supported integral currents. We use Euclidean comass to define current mass, denoted by $\mathbf{M}$. For a compactly supported integral d-current T in $\mathbb{R}^{d+1}$, set

$$\mathbb{F}(T) = \inf\{\mathbf{M}(X) + \mathbf{M}(S) : T = X + \partial S,\ X \in \mathcal{N}_d^c(\mathbb{R}^{d+1}),\ S \in \mathcal{N}_{d+1}^c(\mathbb{R}^{d+1})\}. \tag{1}$$

Integral currents describe oriented surfaces with integer multiplicities. An integral optimum would retain that interpretation in the two parts of the decomposition. Minimization over the integral class alone does not determine whether its value equals the real infimum in (1).

Ibrahim, Krishnamoorthy and Vixie [1] prove this equality and integral attainment for one-currents in the plane (Corollary 3.8). Their introduction also records the case of codimension-one boundaries. Their general codimension-one theorem (Theorem 3.7) uses a subdivision hypothesis to transfer an integral simplicial optimum to the continuous problem. We give a direct argument using scalar BV functions, which does not require that hypothesis.

Theorem 1. *For every integer $d \geq 1$ and every compactly supported integral d-current T in $\mathbb{R}^{d+1}$, there exist $X_I \in \mathcal{I}_d^c(\mathbb{R}^{d+1})$ and $S_I \in \mathcal{I}_{d+1}^c(\mathbb{R}^{d+1})$ such that*

$$T = X_I + \partial S_I, \qquad \mathbb{F}(T) = \mathbf{M}(X_I) + \mathbf{M}(S_I).$$

Here ∂ is the current boundary, $\mathbf{M}$ is mass, and $\mathbb{F}$ is the real infimum (1). The input may satisfy $\partial T \neq 0$. The conclusion asserts existence of an integral optimum and does not require every real optimum to be integral.

The argument rounds a top-dimensional filling rather than approximating the input by a triangulation. Off the input surface, the BV coarea formula controls the variation created by rounding. On the surface, both one-sided values must use the same shift: their rounded difference lies between two adjacent integers. Since the input multiplicity is an integer, the absolute residual is affine on that interval. This gives the cost estimate needed for Theorem 1.

2 Currents and scalar BV functions

Put $n = d+1$ and $\mathrm{vol} = dx_1 \wedge \cdots \wedge dx_n$. For $z \in C_c^\infty(\mathbb{R}^n;\mathbb{R}^n)$, write $\omega_z = \iota_z \mathrm{vol}$. Every finite-mass $(n-1)$-current A has a vector Radon measure μ_A with

$$A(\omega_z) = -\int z \cdot d\mu_A, \qquad \mathbf{M}(A) = |\mu_A|(\mathbb{R}^n). \tag{2}$$

The sign is chosen for the identity below; the mass identity follows from the isometry between Euclidean vectors and $(n-1)$-forms. For $u \in L^1(\mathbb{R}^n)$, define the n-current $[u]$ by $[u](f\mathrm{vol}) = \int uf\,dx$, $f \in C_c^\infty(\mathbb{R}^n)$. If $u \in \mathrm{BV}(\mathbb{R}^n)$, integration by parts gives

$$\mu_{\partial[u]} = Du, \qquad \mathbf{M}([u]) = \|u\|_1, \qquad \mathbf{M}(\partial[u]) = |Du|(\mathbb{R}^n). \tag{3}$$

The derivative Du is distributional; its total variation is the BV seminorm. We always choose Borel representatives of BV functions.

Lemma 2. *Every compactly supported normal n-current in $\mathbb{R}^n$ is $[u]$ for a compactly supported $u \in \mathrm{BV}(\mathbb{R}^n)$.*

Proof. The finite-mass measure representation gives a finite signed measure σ on n-forms. Finite boundary mass means that the distributional gradient $D\sigma$ is a finite vector measure, by testing against ω_z. For nonnegative smooth mollifiers ρ_ε, let $u_\varepsilon = \sigma * \rho_\varepsilon$. Convolution gives

$$\|u_\varepsilon\|_1 \le |\sigma|(\mathbb{R}^n), \qquad \int |\nabla u_\varepsilon|\,dx \le |D\sigma|(\mathbb{R}^n).$$

For $\varepsilon < 1$ the supports lie in one bounded set. BV compactness [3, slides 29–33] gives an L^1-convergent subsequence with limit $u \in \mathrm{BV}(\mathbb{R}^n)$. On the other hand, mollification converges distributionally to σ, so $\sigma = u\mathcal{L}^n$. Testing outside $\operatorname{spt}\sigma$ shows that u vanishes there. The measure representation used here and in (2) is recalled in [3, slides 15–17]. □

For the integral input, (2) takes the form

$$\mu_T = m\nu\,\mathcal{H}^{n-1} \llcorner M, \tag{4}$$

where M is a Borel countably $(n-1)$-rectifiable carrier, ν is a measurable unit normal, and $m \in L^1(\mathcal{H}^{n-1} \llcorner M; \mathbb{Z} \setminus \{0\})$. The sign of m incorporates the chosen orientation. In particular $\mathcal{H}^{n-1}(M) \le \mathbf{M}(T) < \infty$.

We use two classical BV formulas, in the complete forms stated in [2, Theorems A.9 and A.18]. For $u \in \mathrm{BV}(\mathbb{R}^n)$, localized coarea gives

$$|Du|(B) = \int_{\mathbb{R}} |D\mathbf{1}_{\{u>t\}}|(B)\,dt \tag{5}$$

for every Borel B. On an oriented countably $(n-1)$-rectifiable set M, the one-sided approximate L^1 limits u_M^+, u_M^- exist and are finite at $\mathcal{H}^{n-1}$-almost every point, and

$$Du \llcorner M = (u_M^+ - u_M^-)\nu\,\mathcal{H}^{n-1} \llcorner M. \tag{6}$$

The vector-valued source statement gives the scalar one by applying it to $(u, 0, \ldots, 0)$ and taking the first component. For compactly supported u and the carrier (4), the source formulas on a bounded open ball apply on any ball containing the supports in its interior. Indicators of nonzero superlevels are constant near that ball's boundary, so extending them by the same constant introduces no boundary measure. Thus (5) also applies to the full-space superlevel perimeters used below.

3 Rounding without increasing the cost

For $0 < \theta < 1$, let $q_\theta(a) = \lfloor a + \theta \rfloor$. For every $a \in \mathbb{R}$, writing $a = j + r$ with $j \in \mathbb{Z}$ and $0 \le r < 1$ shows that $q_\theta(a)$ takes the values $j, j+1$ with probabilities $1 - r, r$. Hence

$$\int_0^1 q_\theta(a)\, d\theta = a, \qquad \int_0^1 |q_\theta(a)|\, d\theta = |a|. \tag{7}$$

For $a, b \in \mathbb{R}$ and $m \in \mathbb{Z}$, we also have

$$\int_0^1 |m - q_\theta(a) + q_\theta(b)|\, d\theta = |m - a + b|. \tag{8}$$

Indeed, if $a - b = j + r$, then $q_\theta(a) - q_\theta(b) = j + \lfloor \{b + \theta\} + r \rfloor$, where braces denote fractional part. The difference lies between the adjacent integers $\lfloor a - b \rfloor$ and $\lceil a - b \rceil$, and its mean is $a - b$. The function $s \mapsto |m - s|$ is affine on that interval because its corner m is an integer. This proves (8), including all signs.

Lemma 3. *Let $\mu = m\nu\mathcal{H}^{n-1} \llcorner M$ be as in (4), and let $u \in \mathrm{BV}(\mathbb{R}^n)$ vanish almost everywhere off a compact set K. There is a compactly supported $v \in \mathrm{BV}(\mathbb{R}^n; \mathbb{Z})$ such that*

$$|\mu - Dv|(\mathbb{R}^n) + \|v\|_1 \le |\mu - Du|(\mathbb{R}^n) + \|u\|_1. \tag{9}$$

Proof. Take $u = 0$ outside K and set $v_\theta = q_\theta(u)$. Then $v_\theta = 0$ outside K and $|v_\theta| \le |u| + \mathbf{1}_K$. Tonelli's theorem and (7) give

$$\int_0^1 \|v_\theta\|_1\, d\theta = \|u\|_1. \tag{10}$$

For $k \in \mathbb{Z}$, put $E_{k,\theta} = \{u > k - \theta\}$ and define

$$A_\theta(B) = \sum_{k \in \mathbb{Z}} |D\mathbf{1}_{E_{k,\theta}}|(B).$$

Coarea and the partition of $\mathbb{R}$ by the intervals $(k-1, k)$ yield

$$\int_0^1 A_\theta(B)\, d\theta = |Du|(B) \tag{11}$$

for every Borel B. For almost every θ, every sampled level has finite perimeter and $A_\theta(\mathbb{R}^n) < \infty$. Fubini also gives $u(x) + \theta \notin \mathbb{Z}$ for $\mathcal{L}^n$-almost every x, for almost every θ. For these parameters the sums

$$\sum_{k=-N}^{N} \left(\mathbf{1}_{E_{k,\theta}} - \mathbf{1}_{\{0 > k - \theta\}}\right)$$

converge almost everywhere and in L^1 to v_θ. Their absolute values are bounded by $|v_\theta|$. Their derivative series converges in total variation, since its total variation norms sum to $A_\theta(\mathbb{R}^n)$. Distributional testing therefore shows that $v_\theta \in \mathrm{BV}$ and

$$|Dv_\theta| \le A_\theta \quad \text{as measures.} \tag{12}$$

This includes all diffuse variation of u.

Let $a(x) = u_M^+(x)$ and $b(x) = u_M^-(x)$ be the traces in (6). At each point where they exist, the parameters satisfying $a(x) + \theta \in \mathbb{Z}$ or $b(x) + \theta \in \mathbb{Z}$ form a countable set. Since the traces have

measurable representatives and $\mathcal{H}^{n-1}(M) < \infty$, Fubini gives one full-measure set of parameters avoiding these equalities at almost every point of M. For such a parameter, the traces of v_θ are $q_\theta(a)$ and $q_\theta(b)$. To justify this despite the discontinuity of the floor function, if $c + \theta \notin \mathbb{Z}$ choose $\delta > 0$ on which q_θ is constant around c. For all $s \in \mathbb{R}$,

$$|q_\theta(s) - q_\theta(c)| \le (1 + \delta^{-1})|s - c|.$$

Averaging this inequality over a shrinking half-ball proves the assertion from the approximate L^1 trace of u, with no boundedness assumption. Applying (6) to v_θ now gives

$$(\mu - Dv_\theta) \llcorner M = \big(m - q_\theta(a) + q_\theta(b)\big)\nu\mathcal{H}^{n-1} \llcorner M.$$

Consequently (8) implies

$$\int_0^1 |\mu - Dv_\theta|(M)\, d\theta = \int_M |m - a + b|\, d\mathcal{H}^{n-1} = |\mu - Du|(M). \tag{13}$$

On M^c the input measure vanishes, so (12) and (11) give

$$\int_0^1 |\mu - Dv_\theta|(M^c)\, d\theta \le |Du|(M^c).$$

Together with (10) and (13), this bounds the mean total cost by the right-hand side of (9). The cost is measurable: its variation term is the supremum over a countable C^1-dense family of smooth vector tests of supremum norm at most one on a fixed bounded neighbourhood of the supports. Each evaluation is measurable by the joint measurability of v_θ and its uniform L^1 bound. Thus at least one of the common good parameters has cost at most that mean. Choosing it proves the lemma. □

For integer-valued $v \in \mathrm{BV}$ with compact support, $[v]$ has integer density on $\mathbb{R}^n$, finite mass $\|v\|_1$, and finite boundary mass $|Dv|(\mathbb{R}^n)$. The classical boundary rectifiability theorem therefore makes $[v]$ integral; see [4, Theorem 1.1] for an explicit statement and proof, and [1, Section 1.1]. It follows that $T - \partial[v]$ is integral too, and its boundary is ∂T. By (3), Lemma 3 therefore rounds any real admissible decomposition to an integral one without increasing its cost.

4 Attainment with compact support

Proof of Theorem 1. Choose $r_0 > 0$ with $\operatorname{spt} T \subset \operatorname{int} B_{r_0}$. We first confine real competitors to one fixed ball. Take a smooth $\rho : [0, \infty) \to [0, 1]$ that is one up to r_0 and zero after $r_0 + 1$, and let $g(r) = \int_0^r \rho(t)\, dt$. The smooth map

$$f(x) = g(|x|)\frac{x}{|x|} \quad (x \neq 0), \qquad f(0) = 0,$$

is the identity on B_{r_0} and has image in B_{r_0+1}. Its radial and tangential derivative eigenvalues are $g'(r)$ and $g(r)/r$, both in $[0, 1]$. Thus $\|Df\| \le 1$. For compactly supported currents, define $f_\#$ by smooth pullback, using a cutoff equal to one near the current and its boundary. Pullback commutes with exterior differentiation there, so $\partial f_\# S = f_\# \partial S$. The derivative bound and the mass definition give $\mathbf{M}(f_\# A) \le \mathbf{M}(A)$ in every degree. Since $f_\# T = T$, this operation preserves the decomposition and decreases both cost terms. Global properness of f is unnecessary for these compactly supported inputs.

The trivial competitor $(T, 0)$ gives $0 \le \mathbb{F}(T) \le \mathbf{M}(T)$. Take a real minimizing sequence with costs at most $\mathbb{F}(T) + 1/j$, confine it by $f_{\#}$, and write its fillings as $[u_j]$ by Lemma 2. Apply Lemma 3 to choose integer-valued $v_j \in \mathrm{BV}$, supported in the same fixed ball, and put $X_j = T - \partial[v_j]$. Both currents are integral and

$$\mathbf{M}(X_j) + \|v_j\|_1 \le \mathbb{F}(T) + 1/j.$$

Their BV norms are bounded because

$$\|v_j\|_1 \le \mathbb{F}(T) + 1, \qquad |Dv_j|(\mathbb{R}^n) \le \mathbf{M}(T) + \mathbf{M}(X_j) \le \mathbf{M}(T) + \mathbb{F}(T) + 1.$$

BV compactness gives a subsequence converging locally in L^1 to $v \in \mathrm{BV}(\mathbb{R}^n)$. The common support makes convergence global in L^1 and keeps v supported in the closed ball. Pass to a further subsequence with $\sum_j \|v_j - v\|_1 < \infty$. Tonelli then gives pointwise convergence almost everywhere, so v is integer valued.

Set $S_I = [v]$ and $X_I = T - \partial[v]$. The preceding boundary rectifiability argument makes both currents integral and compactly supported. Global L^1 convergence gives $\|v_j\|_1 \to \|v\|_1$ and current convergence $X_j \to X_I$. Mass is lower semicontinuous under current convergence by its supremum definition. Hence

$$\mathbf{M}(X_I) + \mathbf{M}(S_I) \le \liminf_j \left(\mathbf{M}(X_j) + \|v_j\|_1\right) = \mathbb{F}(T).$$

The reverse inequality holds because this pair is admissible for the real infimum. The resulting equality proves the theorem, with $\partial X_I = \partial T$ throughout. □

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